electronics

MDPI

Review

Multi-Dimensional Research and Progress in Parking Space Detection Techniques

Xi Wang, Haotian Miao *, Jiaxin Liang, Kai Li, Jianheng Tan, Rui Luo and Yueqiu Jiang *

College of Information Science and Engineering, Shenyang Ligong University, Shenyang 110159, China
* Correspondence: mht@sylu.edu.cn (H.M.); yueqiujiang@sylu.edu.cn (Y.J.)

Abstract: Due to the increase in the number of vehicles and the complexity of parking spaces, parking space detection technology has emerged. It is capable of automatically identifying vacant parking spaces in parking lots or on streets, and delivering this information to drivers or parking management systems in real time, which has a significant impact on improving urban parking efficiency, alleviating traffic congestion, optimizing driving experience, and promoting the development of intelligent transportation systems. This paper firstly describes the research significance of parking space detection technology and its research background, and then systematically reviews different types of parking spaces and detection technologies, covering a variety of technical means such as ultrasonic sensors, infrared sensors, magnetic sensors, other sensors, methods based on traditional computer vision, and methods based on deep learning. At the end of the paper, the article summarizes the current research progress in parking space detection technology, analyzes the existing challenges, and provides an outlook on future research directions.

Keywords: parking space detection; sensors; computer vision; deep learning

check for updates

Academic Editor: Valentina E. Balas

Received: 6 January 2025
Revised: 5 February 2025
Accepted: 12 February 2025
Published: 14 February 2025

Citation: Wang, X.; Miao, H.; Liang, J.; Li, K.; Tan, J.; Luo, R.; Jiang, Y. Multi-Dimensional Research and Progress in Parking Space Detection Techniques. *Electronics* **2025**, *14*, 748. https://doi.org/10.3390/electronics14040748

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Research on parking space detection technology is of great significance in alleviating urban parking problems, enhancing transportation efficiency and promoting the development of intelligent transportation systems. Such technology helps drivers quickly locate vacant parking spaces by monitoring the status of parking spaces in real time, greatly reducing the time spent on searching and thus effectively alleviating traffic congestion [1]. In addition, this technology can optimize the allocation of parking resources, improve the utilization rate of parking spaces, reduce the waste of resources, and provide data support for urban management to enhance the urban image [2]. At the same time, the development of parking space detection technology also promotes innovation in the fields of sensors, image processing and machine learning, and other technologies, bringing more convenient parking services to drivers and contributing to environmental protection. Therefore, the research and application of parking space detection technology are key factors in realizing urban intelligence and improving the quality of life of residents.

However, the development and implementation of parking detection technology, which is the key to alleviating parking pressure in cities, is encountering a number of challenges. These challenges include, among others, the following:

1. Accuracy enhancement: Ensuring that the parking space detection system is able to recognize available parking spaces with a high degree of accuracy is a top priority in the development of the technology.

2. Environmental adaptability enhancement: Improving the robustness of the system so that it can operate reliably under variable light and climate conditions.
3. Economic benefit assessment: Analyzing the economic feasibility of the technology in order to achieve the best balance between cost and benefit.
4. Universal expansion: Developing solutions that can be widely applied to all types of parking environments to enhance the broad applicability and flexibility of the technology.

Of particular interest is that with the development of high-end computing technologies such as deep learning, their application in parking space detection systems has become a popular research topic. The remarkable progress of deep learning in image and pattern recognition has opened up new horizons for parking space detection technology. However, it also leads to new technical challenges such as how to optimize the performance of algorithms and improve computational efficiency [3]. The purpose of this paper is to provide a comprehensive overview that delves into the development of parking space detection technologies in recent years, listing the different parking space types and mainstream parking space detection technologies, and the challenges they face, as well as an outlook on future research directions.

2. Parking Space Detection Technology Overview

2.1. Type of Parking Space

Parking spaces are categorized in a variety of ways, mainly to accommodate different parking needs, management strategies, and urban planning requirements [4]. With the increasing size of modern vehicles and the complexity of traffic flows, parking space design must now take into account various factors such as vehicle type, operational requirements, and parking angles to optimize space utilization and parking efficiency. For example, different types of vehicles have varying area requirements for parking spaces. A passenger car may require 28.7 square meters, while a 16.5-meter-long road train needs as much as 143.1 square meters. These findings provide a scientific basis for parking space design and highlight the importance of optimizing parking layouts according to different scenarios [5]. This paper summarizes the types of parking spaces, which are divided into the following three main categories based on the type of parking and space utilization:

1. Parallel parking spaces: These spaces are designed to require vehicles to be parked in the direction of the travel lane [6], as shown in Figure 1; they are commonly found on city streets and in space-constrained areas.
2. Vertical parking spaces: As opposed to parallel parking spaces, vertical parking spaces require vehicles to form a perpendicular angle to the travel lane [7], as shown in Figure 2; this design is more common in shopping centers and large parking lots as it facilitates the quick parking and driving away of vehicles.
3. Ramped parking spaces: Ramped parking combines the characteristics of parallel and perpendicular parking; vehicles are parked at a ramped angle. Due to the uncertainty of the angle, there are more forms of this kind of parking [8]—one of them is shown in Figure 3. This design effectively improves space utilization and is particularly suitable for places with high requirements for parking efficiency.

Each of these parking space types has its own characteristics and poses different challenges and requirements for parking space detection techniques, which further raises the demand for the need for adequate generalization capabilities and the adaptability of parking space detection methods.

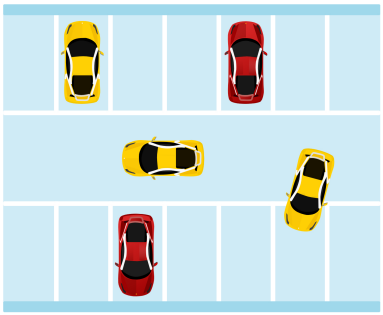


Figure 1. Parallel parking spaces.

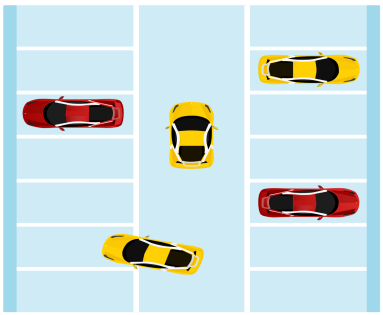


Figure 2. Vertical parking spaces.

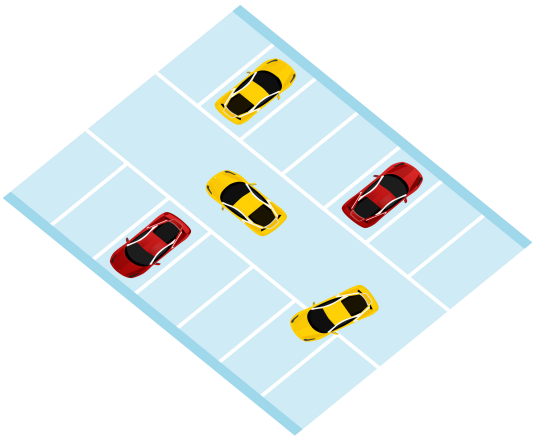


Figure 3. Ramped parking spaces.

2.2. Overview of Detection Technologies

2.2.1. Technical Classification

In recent years, technology has been upgraded as it continues to advance. Technologies for parking space detection are broadly categorized into the following three broad categories:

1. Sensor-based detection technology: Sensor technology recognizes the status of a parking space through direct physical detection, which has the advantage of providing immediate feedback [9]. This technology relies on the stability and cost-effectiveness of the hardware, giving it a significant advantage in real-time monitoring and data collection.
2. Detection techniques based on traditional computer vision: Computer vision techniques focus primarily on the use of image processing algorithms to analyze and identify parking spaces. It relies on feature extraction from images and emphasizes the algorithm's sensitivity to image details and recognition capabilities [10]. This technique demonstrates its flexibility and interpretability in terms of image feature dependencies.

3. Deep learning-based detection techniques: Deep learning techniques automatically learn complex feature representations by training on large amounts of data. They show higher accuracy and automation in dealing with large-scale datasets and complex scenarios, which has led to the increasing application of deep learning in smart parking systems [11].

2.2.2. Application Scenario

Before implementing parking space detection, technology selection needs to closely consider the specific scenario in which it will be applied. The following is a specific discussion of the environments in which different technologies are applicable:

1. Sensor-based technology: This type of technology is particularly suited to outdoor parking environments. Because sensors can directly sense the presence of a vehicle, they remain stable in changing ambient light conditions and exhibit excellent reliability and low false alarm rates, making them particularly suitable for situations where accuracy is critical [12].
2. Traditional computer vision-based technology: This technology is better suited for indoor or environments with predictable lighting conditions, such as underground parking facilities in shopping centers. Computer vision technology works best when the image quality is high and features are more clearly recognizable; thus, it can perform well in controlled lighting conditions [13].
3. Deep learning-based technology: In urban parking environments, deep learning technology is favored for its powerful data processing capabilities. It is capable of processing complex visual data and is suitable for variable environments and large-scale parking lot landscapes [14].

2.2.3. Considerations for Technology Selection

In the process of selecting parking space detection technology, comprehensive considerations must be made, involving a number of key decision-making factors:

1. Economic considerations: the economy of the technology solution is crucial, which is not only related to the initial acquisition cost, but also includes the subsequent maintenance expenditure.
2. Environmental adaptability: the technical solution must demonstrate good environmental adaptability and be able to operate without obstacles in a variety of light and weather conditions as well as in diverse parking lot layouts and parking space types.
3. System operability: the complexity of the technical solution is also an important point of consideration; it should be easy to operate while providing a friendly user experience, avoiding the design of overly complex and difficult to manage systems.
4. Convenience of maintenance and upgrading: the technical solution should facilitate routine maintenance and necessary upgrades to meet the technical evolution and continuous changes in user needs.

Combining these factors, a comprehensive evaluation system can be constructed to guide the choice of technology. To aid in this evaluation, Table 1 provides a detailed comparison of the key characteristics of sensor-based, traditional computer vision-based, and deep learning-based detection methods. Such an evaluation not only ensures that the technology solution is able to cope with current parking space detection needs, but also guarantees its long-term feasibility, adaptability, and user-friendliness. This careful consideration maximizes the benefits of the technology investment and ensures a long-lasting and efficient solution.

Table 1. Key factors for selecting parking space detection technologies.

Reference Number	Factor
Economic Considerations	Initial acquisition cost and subsequent maintenance expenses. Balancing costs and benefits is crucial.
Environmental Adaptability	Ability to operate reliably under various lighting, weather conditions, and diverse parking lot layouts and space types.
System Operability	Ease of operation and user-friendly experience. Avoids overly complex and difficult-to-manage systems.
Maintenance and Upgradability	Facilitating routine maintenance and necessary upgrades to meet evolving technical needs and user requirements.

3. Sensor-Based Parking Space Detection Method

In the exploration of parking space detection technologies, sensor-based methods have become an indispensable component due to their directness and reliability. Sensors, through the detection of physical signals, are capable of providing real-time feedback on the occupancy status of parking spaces, making them particularly suitable for scenarios where high precision is required. Examples of sensor-based parking space detection are illustrated in Figures 4 and 5. However, different types of sensors vary in terms of performance, cost, and applicability in specific environments. These aspects will be elaborated in the subsequent sections.



Figure 4. Magnetic sensor.

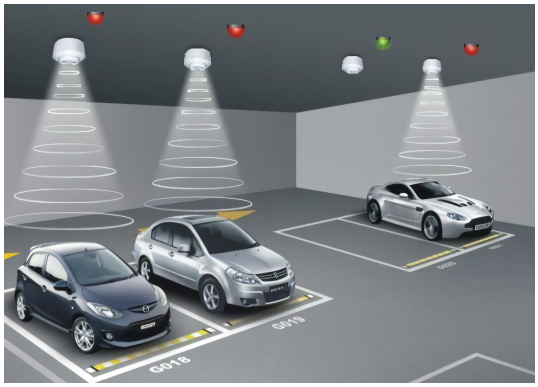


Figure 5. Ultrasonic sensor.

3.1. Ultrasonic Sensors

Ultrasonic sensors play a vital role in parking space detection; when a vehicle passes over these sensors, high-frequency sound waves are emitted which bounce off the sensors.

The sensors can calculate how far away a vehicle is by timing how long it takes for the sound waves to return, providing a cost-effective solution for parking systems [15]. In the invention of Dong Guibin et al. [16], the distance to an obstacle is measured by ultrasonic sensors on the vehicle, data from different sampling points are collected to construct an obstacle contour, and the contour data are optimized by filtering process to accurately identify parking spaces. Although these sensors perform well in terms of cost control and environmental adaptability, and are able to operate independently of lighting conditions, they have some limitations such as limited detection range and susceptibility to interference from surrounding noise. As a result, ultrasonic sensors are usually suitable for small parking lots or parking-intensive areas. In order to attenuate these limitations, other sensors are often fused—e.g., both Haobin Jiang et al. [17] and Piao et al. [18] investigated a data fusion method using ultrasonic sensors and visual sensors to achieve high-precision detection of parking space boundaries, while Yongwang Wang et al. [19] addressed the problem of the automatic parking system to correctly recognize the parking spaces under multiple working conditions, and proposed a method based on a fusion of distance and visual information from camera and ultrasonic sensors for parking space recognition.

3.2. Infrared Sensor

Infrared sensors are commonly used in the development of smart parking systems. Infrared sensors detect obstacles by emitting infrared radiation with high sensitivity, clarity, and precision capture, and have a strong ability to characterize the details of objects [20]; the are also known as general purpose proximity sensors, and they are particularly useful in indoor parking environments because infrared technology is able to work stably in low-light conditions. In an invention by Jinliang Ma [21], an intelligent parking system is introduced, which detects whether a parking space is vacant or not by installing infrared sensors and indicator lights in each parking space and wirelessly transmitting the information to a backend server, enabling fast location of vacant spaces and improving parking efficiency. In the studies of Meenaloshini et al. [22] and Kadir et al. [23], both proposed an IoT-based smart parking management system, where the system utilizes infrared sensors to detect the entry and exit of vehicles and shows the vacant parking spaces to the driver through an LCD screen. The advantages of infrared sensors are their fast response and insensitivity to environmental changes, but they may be interfered by sunlight or other heat sources, limiting their application in outdoor environments. When designing smart parking systems, infrared sensors are often considered for indoor or semi-enclosed spaces to ensure high efficiency and accuracy under changing light conditions.

3.3. Magnetic Sensors

Magnetic sensors detect the occupancy status of a parking space by sensing changes in the magnetic field caused by a vehicle, and they are embedded in the floor of the parking space, providing a covert and stable detection method [24]. Magnetic sensors have the advantages of high sensitivity, small size, flexible installation, and strong anti-interference ability, and are largely unaffected by environmental factors such as light, rain, and snow. A geomagnetic sensor-based vehicle detection algorithm was used in a study by Ying Zhang [25], which proposed a new multi-transient detection algorithm and a W-shaped geomagnetic wireless array layout aimed at improving the accuracy of vehicle detection and reducing the cost of deployment in a large parking lot. Kurek and Macioszek [26] explored the use of magnetic sensors in paid parking zones in the city of Gliwice, Poland, as a means of detecting technology for parking space occupancy. These sensors identify the presence of a vehicle in a parking space by measuring changes in the Earth's magnetic field, and then collect the data through a concentrator and send the information to the controller using a

fiber optic network. Chien et al. [27] used magnetic sensors as a complementary technology aimed at enhancing the system's ability to detect vehicles that are not equipped with Bluetooth beacon emitters as well as solving the problem of vehicles parked across spaces or the system mistakenly identifying moving vehicles as parked vehicles. By integrating magnetic sensors, the accuracy and robustness of the parking management system can be improved to ensure that the parking space status can be effectively monitored in various situations. However, at the same time, magnetic sensors have some limitations; they can be interfered with by the magnetic field of neighboring vehicles. This is especially true for vehicles with high chassis, which have a blind spot for magnetic signals that may not be accurately detected by magnetic sensors; also, the sensitivity of magnetic sensors to changes in the environment may result in unstable readings. Lou et al. [28] proposed an improved magnetic sensor-based vehicle detection method, which fuses the wireless Signal Strength (RSS) to improve the accuracy of detection and optimize the drawbacks of magnetic sensors to some extent.

3.4. Other Sensor Technologies

The choice of parking space detection technologies can be diverse and include pressure sensors, radar sensors, and video sensors, among others. These technologies utilize their own unique physical mechanisms to detect the state of a parking space. For example, pressure sensors rely on changes in the pressure exerted by the vehicle on the ground [29]; radar sensors determine the distance and speed of a vehicle by emitting electromagnetic waves, and most of the current research on vehicle detection by radar is based on LIDAR [30]; and visual sensors rely mainly on image processing and computer vision techniques [31]. Each technology has its unique advantages and limitations, so in practical applications, they are often combined with other technologies to utilize their respective strengths. A SLAM approach combining AVM and LiDAR was proposed in the study by IM et al. [32], which was able to provide a panoramic view of the vehicle's surroundings by utilizing LiDAR's ability to measure accurate distances. This fusion technique not only compensates for the shortcomings of a single sensor, but also enhances the recognition of parking space markers and the accuracy of distance measurement. Sun et al. [33] proposed a multi-sensor fusion method using LiDAR, a gyroscope, and an accelerometer for detecting parking spaces in parking lots. The method collects data through a mobile platform, pre-processes it for feature extraction and pattern matching to identify parking space information, then combines gyroscope and accelerometer data to correct the position of the parking spaces, and finally generates a map of empty parking spaces in the parking lot, which is applicable to a variety of parking environments.

4. Parking Space Detection Method Based on Conventional Computer Vision

Traditional computer vision techniques have long been a cornerstone in the field of parking space detection, leveraging image processing algorithms to analyze and identify parking spaces. These methods rely heavily on the extraction of specific features from images, such as edges, corners, and lines, and are particularly effective in controlled environments where image quality is high. In this section, we delve into several key traditional computer vision methods, including edge detection, corner point detection, line detection, template matching, and feature matching.

4.1. Edge Detection

Edge detection is a key technique in image processing and computer vision that is used to recognize the boundaries and contours of objects in an image. The task of edge detection

is to recognize pixel points with significant changes in brightness as target edges. These edges may represent significant changes in the contours, texture transitions, or surface orientation of an object [34]. For parking space detection systems, it enables the system to recognize parking lines and vehicle contours. Jiang Shaokang [35] used the Canny algorithm to extract the number of edges within a parking space and thus assist in determining the occupancy of parking spaces. Although traditional edge detection algorithms are popular for their accuracy in image boundary recognition, they have limitations in noise processing and complex background recognition, which may result in false detections. Efforts have been made for this purpose; Zhang and Du [36] in their study proposed an image-based approach for detecting parking locations in parking lot environments with special focus on occlusion processing. They used the Holistic Nested Edge Detection (HED) algorithm, which highlights the outer boundaries of an object while attenuating the inner edges of the object and reducing the background noise. Compared to the Canny edge detection method, HED demonstrates superior performance in vehicle edge detection, effectively enhancing the external contours of vehicles and reducing the interference of internal edges and background noise. This research contributes to edge detection and promotes advances related to its development.

4.2. Corner Point Detection

Corner points contain important information about the image; this method is also known as feature points and corner detection and is an important method for extracting image features in machine vision systems [37]. It is particularly important in vision-based parking space detection systems. The corner points of a parking space are the vertices at the entrance of the parking space, and these corner points usually show obvious geometric features in the image, which can be recognized by image processing and computer vision techniques. Commonly used corner detection algorithms include the following: Harris corner detection, Shi–Tomasi corner detection, and FAST corner detector, which are used in specific applications. In a study by Chih-Ming Hsu and Jian-Yu Chen [38], the FAST corner point detector was used to help construct the entire parking space. By detecting the intersections between the parking space boundary lines and the divider lines, these corner points were utilized to identify the structure of the parking space. It was noted that the FAST detector is an accelerated method that significantly improves the detection speed by quickly determining the corner points by examining only four pixels (four compass directions) around a specific pixel in the mask. On the other hand, Eunho Zhan [39] designed a new classification rule for parking space corners which utilizes the YOLOv3 network model for corner detection so as to comprehensively extract parking space information. The method not only determines the location coordinates of the corner points of a parking space, but also obtains the information needed for reasoning about the tail vertices of the parking space, which realizes the precise positioning of the parking space in the surround-view parking image. Corner point detection is useful for understanding the scene structure, but may require high computational resources and may not be stable enough in case of view angle changes or occlusion.

4.3. Line Detection

Line detection is a fundamental task in image processing and computer vision that involves recognizing and extracting line features in an image. These line features can be edges, contours, or human-drawn markers of objects in the image [40]. Radon change is the more classical method for line detection; it is an integral transform used to transform a 2D image into Radon space and detects straight lines and curves in the image by the integral value of that space. Zhu Wangwang et al. [41] proposed an image preprocessing

method based on top-hat transform fused with histogram equalization, which optimizes the detection of parking space lines through Radon transform and threshold segmentation and achieves pixel-level localization of parking space lines through K-Means clustering analysis with non-extremely large suppression processing. The Hough transform outperforms the Radon transform in terms of speed and accuracy, which makes it a preferred technique for detecting lines and curves in images. In the study by Rahman et al. [42], the researchers used the feature extraction method and Hough transform algorithm to identify circular signs in the image which indicate available parking locations. Lu Yibin [43] applied the Hough transform to map the edge points in the image space to the parameter space, and searched for the peaks in the parameter space, which corresponded to the straight lines in the image through the accumulator method; the parameters of the straight lines were extracted based on the peaks in the parameter space, and the boundaries of the parking spaces were plotted in the original image to achieve the detection of the parking spaces. The LSD algorithm is an improved algorithm for the detection of straight lines and has been recognized for its high efficiency and accuracy and attention; it can quickly detect straight line segments in the image and has good robustness to noise and other disturbances in images. Yan Yan [44] proposed a parking space line and corner point recognition method based on an improved LSD algorithm and the mathematical principle of straight-line intersection for improving the accuracy and robustness of target empty parking space detection in automatic parking systems. Future research may focus on further improving the accuracy and speed of line detection techniques, as well as exploring new algorithms and techniques for more complex parking space detection challenges.

4.4. Template Matching

Template matching is an important method in the field of computer vision for recognizing and understanding target images, which recognizes parking spaces with specific shapes or markings by comparing known template images with real-time captured images [45]. Ying Jie et al. [46] proposed an algorithm based on affine transform and template matching for detecting the markings at the entrance of parking spaces and marking them with red circles, which can be effectively recognized even when the markings are partially missing, worn out, or obscured. Yunpeng Wang et al. [47] proposed a method for recognizing vacant parking spaces in outdoor parking lots based on aerial images, where a template image of a vacant parking space is first extracted from an aerial image and used as an ROI, and then the template image is slid over the input aerial image using the sliding window method, and the vacant parking space is located by calculating the matching value between the template image and the corresponding region in the aerial image. In the study by Shidian Ma et al. [48], they proposed a parking space recognition method based on parking space feature construction that uses a template matching algorithm to identify QR code features that may be worn out or contaminated due to vehicle traffic. In the study, they devised a new similarity formula to recognize these features by means of a template matching algorithm, which is effective even when the features are partially damaged or deformed. In addition, they proposed a template matching-based method for parking space angle recognition, which requires that the angle markers of the parking space be intact; if the angle markers are worn out, the recognition will be affected. The accuracy of template matching is highly dependent on the similarity between the template and the actual scene. However, template matching techniques also have some limitations, especially in terms of scalability in large parking areas. In large parking areas with diverse layouts and complex lighting conditions, maintaining high matching accuracy is challenging due to potential template deformation, occlusion, or changes in lighting. Additionally, the computational cost of template matching can be high, especially when processing high-resolution

images or a large number of templates, which may reduce processing speed and affect real-time performance.

4.5. Feature Match

Feature matching techniques have been proposed to make the matching process satisfy certain noise immunity requirements and reduce the ambiguity problem. Generally, the features used for matching should satisfy the following properties: uniqueness, reproducibility, and physical significance. The features used as matching primitives are divided into two categories: local features and global features. Local features include points, edges, line segments, small surfaces, or local energies. Global features include polygons and image structures, etc. [49]. They capture the key features in an image through well-designed descriptors and matches them between different images. Hanbiao Xiao et al. [50] proposed a feature matching method based on binocular inverse projection transformation, which generates new views by inverse projection processing of images captured by binocular cameras, and computes the difference maps of these views to distinguish the ground from the obstacles, thus realizing the fast detection of parking space occupancy. This technique is particularly good at recognizing parking spaces under changing viewing angles and lighting conditions, although it may require more computational resources and has some dependence on the initially selected feature points. In this regard, Zizhang Wu et al. [51] proposed an efficient and generalized parking space detection method called PSDet, which is based on a feature matching technique and improves the detection of parking space vertices by introducing a novel circular descriptor. PSDet employs a two-phase depth architecture, which first coarsely locates the positions of the marked points, and then builds on this with further refinement to improve the detection accuracy. The method demonstrated excellent real-time performance and high accuracy in the experiments, while achieving fast operation speed on the CPU platform, which makes it ideal for application on mobile devices with limited computational resources. In addition, to improve the generalization ability of the feature matching technique for real-world applications, the authors released a large-scale benchmark dataset, PSDD, which covers a wide range of real-world parking scenarios, including brick, grass, diagonal, trapezoidal, open, rectangular, and cubic parking spaces. The release of this dataset provides a valuable resource for future research on enhancing parking space detection in more complex scenarios.

4.6. Contextual Information-Based Approach

Contextual information-based methods utilize environmental information in an image to assist in recognizing and parsing image content. The core idea of this approach is that objects in an image do not exist in isolation, but have a certain relationship with their surroundings [52]. For detecting parking spaces, it improves the accuracy of detection by utilizing clues such as the relative position and orientation of the parking space in the scene. The strength of this technique is reflected in its global approach to consideration; instead of evaluating each parking space individually, it analyzes the parking space as part of a broader scene, effectively reducing the chance of misclassification. In Encío's study [53], a multi-branch output neural network was used so that each branch independently processes the shared feature maps and utilizes all the available contextual information to predict the occupancy of a specific parking space, resulting in more accurate occupancy status predictions in complex visual environments. Do Hoseok and Jin Young Choi [54] proposed a context-based information-based parking space detection method, which mimics the process of finding a parking space by a human driver. Their method consists of two deep network modules: a Parking Context Recognizer (PCR) and a Parking Slot Detector (PSD). The PCR is responsible for recognizing the parking context (type,

angle, and availability of parking spaces), while the PSD utilizes the PCR-obtained parking information obtained by PCR to locate the exact position of the parking space. Contextual information-based approaches, however, may encounter challenges when encountering dynamically changing environments or structurally complex parking facilities. In these contexts, the system may need to undergo a more complex learning process in order to adapt to the changing environment.

4.7. Based on Machine Learning

Machine learning-based approaches recognize and learn the features of parking spaces by analyzing huge image datasets, which in turn enables automatic detection of parking spaces. The workflow starts with the collection and annotation of image data, followed by the identification and extraction of key features in the images [55]. Based on this, appropriate machine learning models, such as support vector machines (SVMs) or neural networks, are selected to process these features and learn them. For example, Gutmann et al. [56] proposed a machine learning model to predict the occupancy of heavy truck parking spaces by fusing Extreme Gradient Boosting (XGBoost) and Long Short-Term Memory Networks (LSTMs) with the help of a model fusion technique of feed-forward neural networks. Gomari et al.'s study [57] used various machine learning algorithms including XGBoost, Random Forest, and Decision Tree to train the model and also considered different feature combinations and algorithms to find the optimal model configuration. Ionita et al. [58] utilized the parking data collected from the SFpark project and the city data from OpenStreetMap by gathering the parking data of the project and using machine learning models such as decision tree, SVM, multilayer perceptron, and gradient boosting tree for feature learning and model training to predict parking occupancy in monitored areas; the study also used cosine similarity and Earth mover distance as similarity metrics to generalize the model to unmonitored areas to achieve accurate prediction of parking occupancy across the city. Li Wang [59] used SVM as a classifier combined with extracted $GLCM_HOG$ fusion feature vectors to classify parking space image blocks for indoor parking space detection. In the study by Tianting Chen and Qing Li [60], a parking space detection and tracking algorithm based on the surround view system was proposed, utilizing Histogram of Orientation Gradient (HOG) features and SVM classifiers for detecting "T"0- and "L"-shaped parking space corners. After sufficient training, the model learns to recognize parking spaces in images and provides real-time detection and status updates in real applications. The strength of this technique lies in its excellent generalization ability and adaptability to changing parking lot environments and conditions. However, it also has some challenges; it needs to be supported by a large amount of data and computational resources, and it requires good real-time processing capabilities. Nevertheless, with the continuous progress of technology, the application of machine learning-based parking space detection methods in the fields of intelligent transportation systems, parking lot management, and automated driving technology is promising and has great potential for development.

In summary, traditional computer vision methods offer a diverse range of solutions for parking space detection, each with its own unique strengths and limitations. Edge detection, corner detection, and line detection excel in identifying the boundaries of parking spaces, while template matching and feature matching enhance the recognition of specific patterns and features. Contextual information-based approaches and machine learning techniques further improve detection accuracy by integrating environmental context and automated feature extraction. These methods are particularly effective in controlled environments but may encounter challenges in dynamic or complex scenarios. Table 2 provides a comparative overview of these methods to offer a clearer understanding of their respective advantages and disadvantages.

Table 2. Comparison of traditional computer vision methods for parking space detection.

Method Type	Advantages	Disadvantages	Applicable Scenarios
Edge Detection	Simple and efficient; suitable for scenes with clear boundaries	Sensitive to noise and complex backgrounds	Indoor parking lots, well-lit conditions
Corner Detection	Provides structural information; suitable for identifying parking space entrances	Computationally intensive; sensitive to occlusions	Indoor parking lots, scenes with clear markings
Line Detection	Accurate in identifying parking lines; robust	Requires preprocessing to handle noise	Indoor parking lots, scenes with distinct markings
Template Matching	Strong recognition ability for specific markings	Sensitive to lighting changes and template deformations	Indoor parking lots, scenes with intact markings
Edge Detection	Simple and efficient; suitable for scenes with clear boundaries	Sensitive to noise and complex backgrounds	Indoor parking lots, well-lit conditions
Feature Matching	High-precision recognition; adaptable to various viewpoints	High computational resource demand; dependent on initial feature points	Indoor parking lots, stable lighting conditions
Contextual Information-Based Methods	Provides holistic scene understanding; reduces misclassification	May fail in dynamic environments or complex structures	Indoor parking lots, simple scene structures
Machine Learning-Based Methods	Automated feature extraction; strong adaptability	Requires substantial data and computational resources	Indoor parking lots, stable lighting conditions

5. Vision-Based Detection Methods Using Deep Learning

In recent years, the rapid development of deep learning has significantly advanced the field of parking space detection. Deep learning-based methods leverage powerful neural network architectures to automatically learn complex feature representations from large-scale datasets, thereby achieving higher accuracy and robustness compared to traditional computer vision techniques. These methods are particularly effective in handling complex scenarios and large-scale parking environments where traditional methods may struggle due to their reliance on handcrafted features and simpler algorithms. The following sections will delve into several key deep learning-based detection methods, including single-stage detection, multi-stage detection, semantic segmentation, instance segmentation, and keypoint detection, each offering unique advantages and applications in the context of parking space detection.

5.1. Single-Stage Detection

Single-stage object detection algorithms are characterized by their simplicity and efficiency. They directly output detection results without requiring candidate region classification, making them faster and computationally more efficient than other deep learning-based detection algorithms. This has led to their widespread application in parking space detection. Representative single-stage object detection algorithms include the YOLO (You Only Look Once) series and the SSD (Single-Shot Multibox Detector) series [61]. The YOLO algorithm integrates detection and classification tasks into a single process, featuring a simple model structure and strong real-time performance. It consists of mod-

ules such as input layers, backbone networks, neck networks, and output layers. From YOLOv1 to YOLOv9 and its numerous improved versions, this algorithm has been widely adopted across various domains [62]. The YOLO algorithm has evolved significantly over the years, with each version offering improvements in accuracy, speed, and efficiency. Table 3 provides a comparative overview of different YOLO versions, highlighting their key features and performance metrics. Li et al. [63] applied a YOLOv3-based detector to simultaneously detect parking slot heads and markers, then inferred the complete parking slots using prior geometric information. Xiao et al. [64] employed an improved YOLOv4 neural network with four fisheye cameras mounted on vehicles to recognize parking spaces. These cameras provided a 360-degree panoramic view, which was processed using inverse perspective mapping to generate bird's-eye views for analysis. The adapted YOLOv4 network directly detected and classified various parking space types, including regular slots, disabled parking spaces, and electric vehicle charging spaces. Zhang Bingli et al. [65] developed a YOLO–Parking algorithm for perception tasks in automated parking systems, using a YOLOv4-based end-to-end multi-task network model. This algorithm utilized surround-view images as input, extracted features via a shared convolutional network, and applied YOLO and DMPR-PS detection heads for parallel detection of parking spaces and obstacles. SSD is another powerful object detection algorithm known for its real-time performance and high detection accuracy [66]. It predicts targets of different sizes using feature maps at multiple scales and supports end-to-end training. Zhang Baojun et al. [67] proposed an improved SSD model to enhance corner detection accuracy in parking spaces, particularly for small targets. By fusing feature maps from different layers and reducing the number of output layers, their model improved detection precision, recall, and F1 scores for L-shaped and T-shaped parking space corners in various environments. Similarly, Li Haitao et al. [68] enhanced the original SSD model by replacing the VGG backbone with a deep residual network and incorporating data augmentation techniques. This improved the accuracy and robustness of parking space detection, making it more suitable for intelligent parking systems. Other single-stage detection methods also show promise. Min et al. [69] introduced an attention-based graph neural network approach that treated parking slot markers as graph-structured data, aggregating neighborhood information to directly predict parking slots in an end-to-end model without manual post-processing. Suhr and Jung [70] proposed an end-to-end single-stage parking slot detection method that utilized convolutional neural networks (CNNs) to extract both global and local parking slot information, integrating them via non-maximum suppression (NMS). Guo Senjie [71] developed an efficient deep learning model by improving the MobileNet architecture, enabling fast and accurate parking space detection without multi-stage processing. Shin Hyunkyung et al. [72] introduced a novel single-stage parking slot detection algorithm based on convolutional network re-parameterization. This algorithm used a multi-branch structure during training to capture local image details, improving detection precision. Despite potential challenges in detecting small objects or handling complex backgrounds, single-stage detectors hold significant promise for urban parking management, particularly in dynamic environments requiring rapid responses.

Table 3. Comparative overview of different YOLO versions.

Version	Release Year	Main Improvements	mAP@0.5 (%)	FPS	Features	Applications
YOLOv1	2016	Initial version with end-to-end single-stage detection framework	69.5	45	Simple and fast, but weak in small object detection	Real-time video stream detection
YOLOv2	2017	Batch normalization, high-resolution classifier, anchor box clustering	78.6	67	Improved detection accuracy and speed, optimized small object detection	Real-time object detection, security surveillance
YOLOv3	2018	Multi-scale detection, new anchor box design, support for multi-class detection	82.6	50	Multi-scale detection, suitable for complex scenes	Multi-class object detection, autonomous driving
YOLOv4	2020	Weight decay, CSPDarknet53, Mish activation function	83.2	65	Enhanced feature extraction and training efficiency	Real-time surveillance, industrial inspection
YOLOv5	2020	PyTorch framework, automatic anchor box learning, simplified model structure	83.0	120	Easy to deploy and expand, supports multiple input sizes	Real-time detection, lightweight devices
YOLOv6	2022	Further optimized model structure to improve inference speed	83.5	150	Optimized for real-time performance, lightweight design	Mobile devices, edge computing
YOLOv7	2022	Scalable model architecture with trade-offs between accuracy and speed	84.0	180	Enhanced small object detection, model compression	Real-time detection, multi-scene adaptation
YOLOv8	2023	Improved generalization and efficiency based on YOLOv7	84.5	200	Better small object detection, suitable for complex environments	Autonomous driving, intelligent transportation
YOLOv9	2024	Introduced Programmable Gradient Information (PGI), Generalized Efficient Layer Aggregation Network (GELAN), optimized training strategies	84.8	160	High accuracy, low computational complexity, supports multi-tasking (detection, segmentation)	Autonomous driving, video surveillance, robotic vision

5.2. Multi-Stage Detection

Models based on multi-stage detection frameworks typically exhibit higher accuracy than single-stage and two-stage detection frameworks [73]. A prominent example is Faster R-CNN, which achieves higher parking space detection accuracy through refined multi-stage processing. Faster R-CNN replaces the selective search process in Fast R-CNN with a Region Proposal Network (RPN), providing more precise region proposals while reducing computational redundancy and improving detection speed [74]. In parking space detection, multi-stage methods often first identify potential parking areas in images or video streams before precisely determining their boundaries, dimensions, and orientations. Bui and Suhr [75] advanced this field by proposing a two-stage parking slot detection method based on convolutional neural networks (CNNs) with specialized multi-scale feature extraction. This addressed the limitations of general two-stage methods in parking slot detection, enhancing performance. Karanam et al. [76] further refined this approach by employing a multi-camera system to manage parking spaces under real-world conditions, including occlusions, lighting changes, and weather variations, thereby improving system robustness. Huang et al. [77] designed a 360-degree panoramic system combined with a Faster R-CNN model to address issues such as uneven lighting and complex backgrounds. This significantly enhanced detection accuracy. Marek's research [78] introduced a Faster R-CNN FPN model, leveraging feature pyramid networks for multi-scale feature extraction and incorporating known parking coordinates for feature pooling. This effectively improved detection accuracy and generalization capabilities. These studies highlight the evolution

of parking space detection technologies and pave the way for future advancements in this field.

5.3. Semantic Segmentation

Semantic segmentation is a foundational computer vision task that involves pixel-level perception and understanding. Its objective is to assign a corresponding class label to every pixel in an input image [79]. This technique enriches scene understanding for parking space detection, helping systems differentiate parking spaces from other ground markings, and provides detailed environmental information for intelligent parking navigation. While computationally expensive, semantic segmentation holds significant potential in high-precision parking assistance systems.To enhance the accuracy and robustness of parking space detection, Jian and Lin [80] proposed an end-to-end trained model that integrates semantic segmentation of lines and points. Based on multi-task learning, this model generates images of entry lines and corner center points, enabling more precise determination of actual parking space coordinates during the post-processing step. Addressing the challenge of detecting parked vehicles in stereo satellite imagery, Zambanini et al. [81] introduced a novel method utilizing high-resolution stereo satellite images and an adapted Faster R-CNN object detector. This approach effectively mitigates interference from extraneous elements. In the study by Lee et al. [82], semantic segmentation was employed to process image data from surround-view monitoring systems. The system classified each pixel in the image into its corresponding category, enabling not only the detection of parking spaces but also understanding their exact location and boundaries in the image. This provides precise parking space information for automated parking systems. Similarly, Lai et al. [83] proposed a Real-Time Parking Space Detection (RPSD) network leveraging semantic segmentation techniques to improve perception of parking spaces and their surrounding environments. Li Bing [84] developed a lightweight semantic segmentation network, UFAC-Net, which combines depthwise separable convolution and asymmetric convolution strategies. Additionally, a slope-based clustering algorithm for parking line extraction was designed, improving both the precision and speed of parking space detection. However, these methods exhibit certain limitations that warrant further exploration and refinement in future studies.

5.4. Instance Segmentation

Instance segmentation combines the strengths of object detection and semantic segmentation, functioning as a pixel-level task that not only locates all objects in an image but also classifies them at the pixel level. This technique can distinguish between objects of different classes and identify distinct instances within the same class [85,86]. Instance segmentation is particularly significant in parking space detection, as it helps systems differentiate and track visually similar parking spaces, thereby significantly enhancing adaptability and recognition accuracy in complex parking environments. Cheng Yuanhang and Yu Jun [87] proposed a parking space status recognition algorithm based on Mask R-CNN. This algorithm detects vehicles in video images and uses object detection and segmentation to obtain vehicle bounding boxes and coordinate positions. By employing the Intersection over Union (IoU) method, the algorithm determines the overlap between vehicle and parking space bounding boxes, enabling intelligent recognition of parking space status. Although the algorithm exhibits excellent recognition efficiency and accuracy, it may be sensitive to camera angles and lighting conditions.To address these sensitivities, researchers at Institut Teknologi Sepuluh Nopember [88] proposed a preprocessed Mask R-CNN method. Using contrast enhancement techniques, particularly the Exposure Fusion framework, this approach mitigates issues caused by lighting variations in open environ-

ments, significantly improving the accuracy of parking space markings. However, it still faces challenges when handling occlusions. Further advancements were made by Shaokang Jiang et al. [89], who integrated Mask R-CNN with the LSD algorithm. Their approach not only detected parking slot markers but also extracted guiding lines and parallel lines to identify candidate parking spaces. This method demonstrated robustness under complex lighting conditions and varying camera perspectives, accurately detecting multiple types of parking spaces, including perpendicular, parallel, and angled slots. Their research lays a solid foundation for future developments in this field.

5.5. Keypoint Detection

Keypoint detection is a critical technique in computer vision that transforms object detection into a problem of predicting central points. These central points represent the target objects, and their precise locations are obtained through offsets, while regression methods are used to determine object dimensions [90]. This technology has been widely applied in various fields, including but not limited to human pose estimation, facial recognition, gesture recognition, medical image analysis, and autonomous driving. In the context of parking space detection, deep learning-based keypoint detection technologies focus on enhancing detection accuracy and efficiency. Wang Jiazhou and Sang Haifeng [91] developed an end-to-end network that integrates keypoints, local features, and global features using a cross-attention mechanism. Qin Xiaohui et al. [92] proposed an anchor-free end-to-end approach specifically targeting the orientation of parking slot entry lines, simplifying keypoint matching and orientation determination. Shan Kaiqiang and Sang Haifeng [93] introduced a dual-path parallel multi-scale network that improves the accuracy of marker point position and angle detection through feature fusion. Later, Shan Kaiqiang [94] advanced this further by proposing a keypoint-based parking space detection technique using panoramic video. This method captures fisheye camera images around the vehicle to construct panoramic images. A deep learning network, particularly a dual-path multi-resolution detection network, is then used to predict the positions and angles of parking space keypoints. By employing Gaussian heatmaps and keypoint matching strategies, the method reconstructs complete parking spaces while identifying their types and occupancy statuses. Yue Pengyu [95] created the PS-rap dataset, which includes annotations for parking space keypoints, horizontal rectangular boxes, and rotated rectangular boxes. Using an improved CenterNet model, this method detects the center points and four keypoints of parking spaces, subsequently determining their positions, types, and occupancy statuses. Tian Fei et al. [96] developed a parking space detection method based on keypoint instance detection. Their approach utilizes a MobileNet V2 feature extraction network within the detection model. The model's parking classification branch detects parking space center points, while the keypoints regression branch predicts parking space keypoints. By comparing the regressed keypoints with detected keypoints and selecting those with higher confidence scores, the method identifies final parking space instances along with their keypoint coordinates. The application of these techniques has significantly advanced the deployment of autonomous driving and intelligent transportation systems in real-world scenarios. These developments demonstrate the powerful potential of deep learning in keypoint detection, paving the way for more robust and efficient parking space detection solutions.

5.6. Deep Learning Optimization Techniques

Deep learning optimization techniques encompass a range of methods aimed at enhancing the performance of deep neural networks, reducing computational resource consumption, accelerating training, and improving model generalization capabilities [97].

In the field of parking space detection, optimization strategies focus on algorithmic iterations, dataset expansion and augmentation, and overall system performance enhancement. Li Weidong et al. [98] utilized model simplification techniques, selecting MobileNetV3 as the backbone for feature extraction. This streamlined network architecture significantly reduced parameter count and computational cost. Their study also incorporated feature pyramids to capture multi-scale features, improving parking space detection capability. Additionally, the use of multi-task learning allowed simultaneous classification, bounding box localization, and semantic segmentation, fostering a comprehensive understanding of parking space features. In the work of Hayawi and Muhammad [99], a pre-trained ResNet50 model was employed, leveraging residual learning to address vanishing gradient issues in deep networks and optimize feature extraction. The training process utilized optimization algorithms such as Adam to update network weights, achieving high accuracy on both public and local datasets. The system demonstrated strong transfer learning capabilities on unseen images, showcasing robust generalization under various environmental conditions. Nurullayev et al. [100] enhanced feature extraction capabilities by using dilated convolutions, which effectively expanded the receptive field in convolutional operations without increasing the parameter count. This allowed the model to capture more contextual information, improving accuracy in identifying parking space occupancy under diverse environmental conditions. Nyambal and Klein [101] applied efficient optimizers, such as Nesterov Accelerated Gradient (NAG) and Stochastic Gradient Descent (SGD), to enhance training efficiency and accuracy, exemplifying the application of optimization techniques in deep learning. Martynova et al. [102] introduced the EfficientNet-P model, an improved version of the EfficientNet architecture. By integrating techniques such as neural architecture search, compound scaling, depthwise separable convolutions, Swish activation functions, and squeeze-and-excitation blocks, they significantly enhanced parking space detection accuracy and efficiency. In the study by Wang Jixiang and Zhang Aimei [103], transfer learning techniques were applied using the VGG-16 network for parking space state classification. Improved image preprocessing and edge detection techniques further optimized model accuracy, particularly in handling lighting variations and occlusion, demonstrating robustness. Yamin Siddiqui et al. [104] proposed a parking space detection method based on Deep Extreme Learning Machine (DELM). By simplifying the training process and reducing memory usage, DELM achieved high efficiency while significantly lowering training time and complexity compared to traditional artificial neural networks. Yuldashev et al. [105] improved the structure of the MobileNetV3 model by introducing new activation functions, more efficient attention mechanisms, and replacing convolution operations. These enhancements increased the accuracy of parking space occupancy classification. Similarly, Manh [106] introduced variational information bottleneck (VIB) loss to optimize feature extraction, enabling the model to learn more generalized and robust feature representations for effective parking space detection in diverse environments. Amato et al. [107] focused on model miniaturization and real-time performance, designing a compact CNN architecture (mAlexNet) tailored for smart cameras, enabling real-time operations on resource-constrained devices. Their work also introduced the CNRPark-EXT dataset to further assess model performance and generalization in parking occupancy detection tasks. These optimization techniques collectively make deep learning models more flexible and efficient, enabling them to adapt to diverse deployment environments and application needs, driving advancements in intelligent transportation and parking systems.

6. Comparative Study

In Table 4, a comparative study of different parking space detection methods is provided. It includes detailed information such as the model types used in the referenced

studies, datasets utilized, the number of images used by the models, model accuracy, types of models employed, input sizes of the models, and other specific details.

Table 4. Summary of value-based reinforcement learning algorithms.

Reference Number	DataSet	Number of Images Used by Model	Model Accuracy	Model Recall	Model Type	Model Input Size (px)
[35]	Self-collected	6000	93.9%	92.5%	Mask RCNN, LSD line detection	16*16
[39]	PS2.0	9827 (training) 2338 (testing)	97.39%	97.34%	YOLOV3	160*100*3 (head region of parking spaces)
[46]	Self-collected	4492 (1524 sunny, 1481 cloudy, 1487 rainy)	98.58%	-	PPHT and line clustering	-
[48]	Custom QR code	120	95.45%	93.33%	Canny edge detection, Hough transform, template matching algorithm	-
[51]	PSDD	14,628	95.67%	98.21%	PSDet	320*240
[53]	ETSIT, PUCPR, UFPR04, UFPR05	-	>99.1%	-	Multi-branch output ConvNeXt (MBONN)	267*200
[54]	ps2.0	12,165	99.63%	99.31%	YOLOv3 + Custom DCNN model	600*600
[65]	ps2.0	5349 (training), 595 (validation)	93.53%	-	YOLOV4	-
[67]	ps2.0	12,165	88.11%	91.85%	Improved SSD model	300*300*3
[70]	ps2.0	9827 (training), 2338 (validation)	99.77%	99.77%	VGG-16	416*416
[78]	ACPDS (Action-Camera Parking Dataset)	293, 11,236 unique parking slot	98%	-	R-CNN and Faster R-CNN FPN	R-CNN: 64, 128, 256; Faster R-CNN FPN: 800, 1100, 1440
[88]	CNRPark	270	85.8%	-	Mask R-CNN	1920*1080
[89]	ps2.0	9827 (training), 2338 (validation)	Rectangular: 99.65%, Inclined: 99.04%	-	End-to-end trainable network with cross-attention mechanism	3*512*512
[91]	ps2.0	9827 (training), 2338 (validation)	99.77%	99.77%	VGG-16	416*416
[98]	ps2.0	197	98.39%	97.21%	MobileNetV3	512*512
[99]	PKLot and local dataset	PKLot: 12,417, Local: 175	PKLot: 99.67%, local: 99.12%	-	ResNet50	224*224
[105]	CNRPark-EXT and PKLot	CNRPark-EXT: 157,549; PKLot: 695,899	CNRPark_EXT: 98.01%	-	Improved MobileNetV3	224*224

This study provides an in-depth analysis of different parking detection methods. From the results, it can be observed that deep learning models deliver the highest accuracy of 99.77%. Following this, the ResNet50 network achieved an accuracy of 99.67% using a dataset with 12,417 images. Lastly, an accuracy of 99.65% was achieved for the detection of rectangular parking spaces on the PS2.0 dataset.

7. Conclusions and Outlook

7.1. Conclusions

In recent years, with the continuous improvement of drivers' requirements for cars, parking assistance systems have been developed rapidly, and the technology of parking space detection has also been improved. According to the different key technologies of parking space detection, this paper elaborates on the relevant research in the field of parking space detection and analyzes and introduces different parking space detection methods based on sensors, traditional computer vision, and deep learning; information regarding the cited literature is shown in Figure 6. The vast majority of the cited articles for each type were published within the last five years, and the specific proportion is shown in Figure 7.

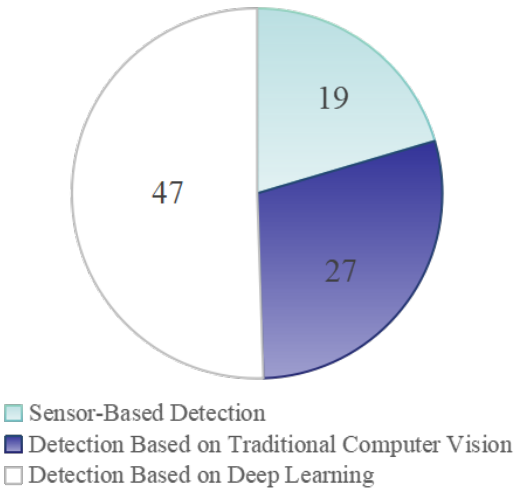


Figure 6. Number of References for Parking Space Detection Technologies.

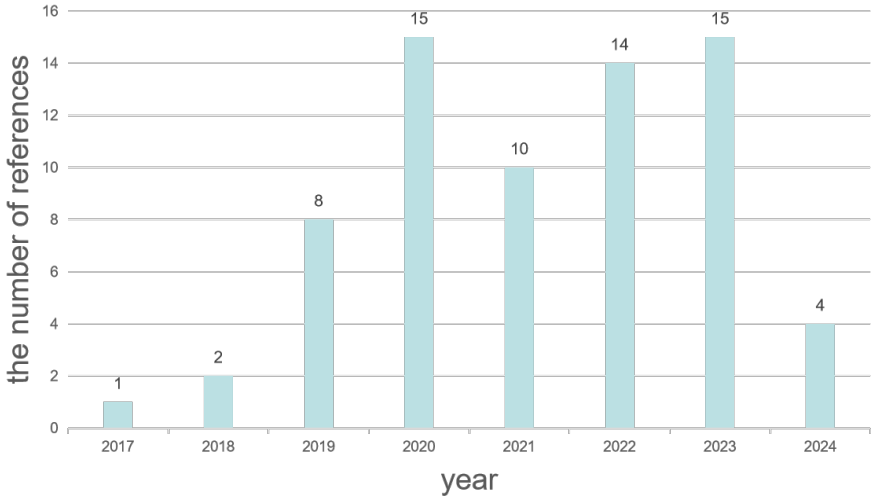


Figure 7. Yearly distribution of references.

7.2. Challenges and Future Directions

Currently, the mainstream approach is deep learning-based parking space detection, which improves the accuracy and speed of recognizing parking spaces by using convolutional neural networks (CNNs). However, current parking space detection technology still faces some challenges. In terms of environmental adaptation, the performance of existing models tends to degrade under different lighting and weather conditions as well as in complex urban contexts, such as occlusion and shadows, and there is a need to further

improve the robustness of current models to these changes. In terms of real-time performance, despite the progress made in the accuracy of deep learning models, achieving fast detection to meet real-time demands while maintaining high accuracy, especially in systems with limited computational resources, remains a technical challenge. The data diversity challenge involves acquiring training data that are rich and representative enough to cover a wide range of parking lot scenes and geographic locations, and while current datasets are diverse, applicability needs to continue to be enhanced. In terms of generalization capability, models tend to perform well in environments with similar training data, but their generalization capability is insufficient when confronted with new or uncommon parking layouts and signs, which limits their performance in unknown environments. In addition, model lightweighting and optimization is also a hot topic in current research, aiming to enable deep learning models to run efficiently on edge devices while maintaining or improving detection performance. Finally, with the development of technology, users' expectations of parking space detection systems are increasing, which requires the systems not only to accurately detect parking spaces, but also to provide more personalized and intelligent services, such as intelligent recommendation of parking spaces based on user behaviors and preferences. Addressing these challenges will be the key to driving parking space detection technology forward. This paper argues that further research can be promoted in the future in the following areas.

1. Multi-Modal Sensor Fusion

Although accuracy and robustness in parking space detection can be enhanced by integrating data from multiple sensors, future research will focus more on the comprehensive use of multi-modal sensor data. This will not only involve traditional sensors like vision sensors, radar, and LiDAR but also data from emerging sources such as IoT devices and in-vehicle information systems. Advanced data fusion algorithms would need to be developed to facilitate deeper feature extraction and comprehensive integration. Challenges such as temporal synchronization, spatial calibration, and improving computational efficiency during the fusion process would also be addressed. These efforts are expected to ensure that parking space detection systems remain stable and reliable under diverse environmental conditions.

2. Optimizing Deep Learning Models

The optimization of deep learning models is considered crucial for advancing parking space detection technologies. Efforts should focus on the development of innovative network architectures and refined parameter compression techniques to create lighter models with stronger generalization capabilities. Compact models capable of operating efficiently on edge devices, even under resource-limited conditions, should be prioritized. Maintaining high detection accuracy under varying lighting conditions, diverse vehicle types, and complex parking layouts is also emphasized. These advancements are projected to significantly support the development of intelligent traffic systems and autonomous driving technologies.

3. Integration of Intelligent Parking Systems

Integrated intelligent parking systems are seen as essential for the growth of smart cities. Future solutions are expected to achieve higher levels of intelligence and interconnectivity. Seamless integration with drivers' smartphones, in-vehicle navigation devices, and other urban smart services is anticipated to deliver personalized parking experiences. Systems are expected to predict parking demand, reserve spaces, and track parking durations and fees automatically. Data sharing and collaboration with other urban infrastructures, such as smart lighting and energy management systems, are also predicted. This integration aims to optimize resource usage and reduce environmental impact. Intelligent parking

systems are projected to become a cornerstone of smart cities, providing more convenient, efficient, and eco-friendly parking experiences.

7.3. Framework for Advancing Parking Space Detection Technology

Throughout this comprehensive review of parking space detection technologies, we have identified several key issues and challenges that arise in the deployment and operation of these systems. Although the previous sections have provided a detailed outlook on future research directions, this section aims to provide a structured framework to address the operational and implementation challenges that arise during the implementation of these technologies. The key components of the framework are as follows.

1. Enhanced System Integration and Interoperability

One of the key challenges in deploying parking space detection systems is the integration with existing infrastructure and other smart city initiatives. Future systems should focus on developing robust interfaces and protocols to enable seamless communication and data sharing between different components. This includes not only the integration of various sensor technologies but also coordination with traffic management systems, payment gateways, and user interfaces.

2. Robust Error Detection and Correction Mechanisms

Despite advancements in detection technology, errors and false alarms remain a significant issue in complex and dynamic environments. Implementing advanced error detection and correction algorithms can significantly enhance the reliability of these systems. This includes real-time monitoring of sensor data, anomaly detection, and adaptive correction mechanisms that can adapt to changing conditions.

3. User-Centric System Design and Feedback Mechanisms

While future research directions emphasize the importance of user experience, this framework highlights the necessity of user feedback mechanisms. Systems should be designed to collect and analyze user feedback to continuously improve performance and user satisfaction. This can include mobile applications that allow users to report inaccurate information, suggest improvements, and provide real-time feedback on system performance.

4. Economic and Environmental Impact Assessment

The deployment of parking space detection systems should not focus solely on technological advancements but also consider their economic feasibility and environmental impact. Future research should include comprehensive cost–benefit analyses and life cycle assessments to ensure that these systems are economically viable and environmentally sustainable. This includes evaluating the energy consumption of sensors, the materials used in system components, and the overall carbon footprint of the technology.

5. Scalability and Flexibility in System Design

To meet the diverse needs of different urban environments, the design of parking space detection systems must be scalable and flexible. This includes developing modular system architectures that can easily adapt to various parking scenarios, from small private lots to large public facilities. Additionally, system design should accommodate future technological upgrades and changes in user requirements.

This framework aims to provide a structured approach to addressing the key issues in the deployment and operation of parking space detection technologies. By focusing on enhanced system integration, robust error detection, user-centric design, economic and environmental impact assessment, and scalable system design, we can ensure that

these technologies are not only innovative but also practical, reliable, and sustainable. This framework complements the future research directions outlined in the previous sections by providing a roadmap to overcome existing challenges, ensuring the successful implementation of these technologies in real-world scenarios.

Author Contributions: X.W. was responsible for material preparation, data collection, and analyses. She also penned the first draft of the manuscript. Subsequently, J.L., K.L. and J.T. undertook the comprehensive review and meticulous editing of the manuscript to enhance its clarity and academic rigor. The project was carried out under the supervision of H.M. and Y.J., who provided guidance and oversight throughout the research process. In addition, R.L., H.M. and Y.J. were in charge of project administration, handling tasks such as resource allocation and scheduling. All authors have read and agreed to the published version of the manuscript.

Funding: This project was funded by the Research Support Program for Introducing High-Level Talents from Shenyang Ligong University (1010147001225), the Ministry of Education's supply–demand docking employment and education project (2023122570529), and Special Funding for Basic Research Operating Costs of Undergraduate Colleges and Universities in Liaoning Province (SYLUGXTD07).

Data Availability Statement: No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Chen, S. Intelligent parking-parking space artificial intelligence detection system. *Inf. Syst. Eng.* **2024**, 27–30. [CrossRef]
2. Li, W.; Cao, H.; Liao, J.; Xia, J.; Cao, L.; Knoll, A. Parking slot detection on around-view images using DCNN. *Front. Neurorobot.* **2020**, *14*, 46. [CrossRef]
3. Wong, G.S.; Goh, K.O.M.; Tee, C.; Md. Sabri, A.Q. Review of vision-based deep learning parking slot detection on surround view images. *Sensors* **2023**, *23*, 6869. [CrossRef]
4. Kumar, K.; Singh, V.; Raja, L.; Bhagirath, S.N. A review of parking slot types and their detection techniques for smart cities. *Smart Cities* **2023**, *6*, 2639–2660. [CrossRef]
5. Mikusova, M.; Abdunazarov, J.; Zukowska, J.; Jagelcak, J. Designing of Parking Spaces on Parking Taking into Account the Parameters of Design Vehicles. *Computation* **2020**, *8*, 71. [CrossRef]
6. Li, Y.; Zu, Y.; Si, Y.; Hu, J. Analysis of the public security industry standard specification for the setting of parking spaces on urban roads (ga/t 850-2021). *China Stand.* **2021**, 186–191. [CrossRef]
7. Ma, X.; Zhou, C.; Gu, J. Effect of vertical parking space width and passage width on drivers' parking behavior. *Smart City* **2018**, *4*, 10–13. [CrossRef]
8. Chen, Z.; Peng, Y.; Huang, W.; Yao, Y.; Wu, Q.; He, W. Parking space selection method under quantitative evaluation of environmental complexity. *J. Fuzhou Univ. (Nat. Sci. Ed.)* **2024**, *52*, 61–68.
9. Paidi, V.; Fleyeh, H.; Håkansson, R.G. Smart parking sensors, technologies and applications for open parking lots: A review. *IET Intell. Transp. Syst.* **2018**, *12*, 735–741. [CrossRef]
10. Ma, Y.; Liu, Y.; Shao, S.; Zhao, J.; Tang, J. Review of research on vision-based parking space detection method. *Int. J. Web Serv. Res. (IJWSR)* **2022**, *19*, 1–25. [CrossRef]
11. Long, J.; Zhou, Y. Real time detection of outdoor parking space based on alexnet neural network. *China Sci.* **2021**, *16*, 295.
12. Zhang, C.; Qin, L.; Yang, H.; Zhao, Z.; Tan, Z. Summary of research progress on Key Technologies of automatic parking. *J. Automot. Eng.* **2023**, *13*, 603–614.
13. Xu, Z.; Tang, X.; Ma, C.; Zhang, R. Research on Parking Space Status Recognition Method Based on Computer Vision. *Sustainability* **2022**, *15*, 107. [CrossRef]
14. Xu, Z.; Tang, X.; Ma, C.; Zhang, R. Research on parking Space Detection and Prediction Model based on CNN-LSTM. *IEEE Access* **2024**, *12*, 30085–30100. [CrossRef]
15. Zhu, Y. Application of ultrasonic sensor in intelligent vehicle system. *Spec. Purp. Veh.* **2023**, 31–33. [CrossRef]
16. Zhuhai Shangfu Electric Technology Co., Ltd. A Parking Space Detection Method and Automatic Parking Method. CN Patent CN202010067655.6, 10 May 2024.
17. Jiang, H.; Ye, H.; Ma, S.; Chen, L. High precision parking space identification method of automatic parking system based on multi-sensor data fusion. *J. Chongqing Univ. Technol. (Nat. Sci.)* **2019**, *33*, 1–10.

18. Piao, C.; Zhang, J.; Chang, K.; Li, Y.; Liu, M. Multi-Sensor Information Ensemble-Based Automatic Parking System for Vehicle Parallel/Nonparallel Initial State. *Sensors* **2021**, *21*, 2261. [CrossRef]
19. Wang, Y.; Wang, S.; Jiang, H.; Zhou, N.; Li, Z. Research on parking space recognition method of automatic parking system under multiple working conditions. *J. Electron. Meas. Instrum.* **2023**, *36*, 174–182.
20. Peng, J.; Li, F.; Xiong, L. Automatic target recognition method in complex scenes based on infrared sensors. *J. Sens. Technol.* **2024**, *37*, 1377–1382.
21. Xuzhou Kangbo Intelligent Control Instrument Co., Ltd. A Parking Lot Vacancy Detection Device and Method. CN Patent CN201911066023.1, 28 February 2020.
22. Meenaloshini, M.; Ilakkiya, J.; Sharmila, P.; Malar, J.S.; Nithyasri, S. Smart car parking system in smart cities using IR. In Proceedings of the 2019 3rd International Conference on Computing and Communications Technologies (ICCCT), Chennai, India, 21–22 February 2019; IEEE: Piscataway, NJ, USA, 2019; pp. 178–182.
23. Abd Kadir, M.M.; Osman, M.N.; Othman, N.A.; Sedek, K.A. IoT based car parking management system using IR sensor. *J. Comput. Res. Innov.* **2020**, *5*, 75–84. [CrossRef]
24. Wang, X.; Wang, F. Development and application of magnetic sensors. *Magn. Mater. Devices* **2020**, *51*, 63–66. [CrossRef]
25. Zhang, Y. Research and Implementation of Vehicle Detection Algorithm Based on Geomagnetic Sensor. Master's Thesis, University of Electronic Science and Technology of China, Chengdu, China, 2023.
26. Kurek, A.; Macioszek, E. Drivers' subjective assessment of the ease of finding a vacant parking space in an area equipped with vehicle detection devices. *Sensors* **2022**, *22*, 6734. [CrossRef] [PubMed]
27. Chien, C.F.; Chen, H.T.; Lin, C.Y. A low-cost on-street parking management system based on bluetooth beacons. *Sensors* **2020**, *20*, 4559. [CrossRef]
28. Lou, L.; Zhang, J.; Xiong, Y.; Jin, Y. An improved roadside parking space occupancy detection method based on magnetic sensors and wireless signal strength. *Sensors* **2019**, *19*, 2348. [CrossRef] [PubMed]
29. Shenzhen Huayang International Engineering Design Co., Ltd. BIM Based Pressure Sensor Parking Space Detection Device. CN Patent CN201520894572.9, 3 August 2016.
30. Zhang, X.; Zhang, Z.; Li, Y.; Guo, X.; Chen, W. Vehicle detection algorithm based on pulse coherent radar. *Comput. Eng. Appl.* **2022**, *58*, 276–282.
31. Zheng, W. Research on Parking Space Recognition Based on Visual Sensor and Deep Learning. Master's Thesis, Harbin University of Science and Technology, Harbin, China, 2021. [CrossRef]
32. Im, G.; Kim, M.; Park, J. Parking line based SLAM approach using AVM/LiDAR sensor fusion for rapid and accurate loop closing and parking space detection. *Sensors* **2019**, *19*, 4811. [CrossRef] [PubMed]
33. Sun, D.; Huang, X.; Hu, X.; Zhang, Y.; Zhao, M. Parking Lot Detection Method Based on Multi-Sensor Information Fusion. CN Patent CN201710431169.6, 24 March 2020.
34. Yang, X.; Jun, Z. Overview of image edge detection. *Comput. Eng. Appl.* **2023**, *59*, 40–54.
35. Jiang, S. Research on Parking Space Intelligent Recognition and Parking Path Tracking Control Based on Look Around System. Master's Thesis, Jiangsu University, Zhenjiang, China, 2021.
36. Zhang, C.; Du, B. Image-based approach for parking-spot detection with occlusion handling. *J. Transp. Eng. Part A Syst.* **2020**, *146*, 04020098. [CrossRef]
37. Shang, S.; Cao, J.; Wang, M.; Zheng, X.; Gao, H. Research on corner detection algorithm in machine vision. *Comput. Meas. Control.* **2024**, *32*, 217–225. [CrossRef]
38. Hsu, C.M.; Chen, J.Y. Around view monitoring-based vacant parking space detection and analysis. *Appl. Sci.* **2019**, *9*, 3403. [CrossRef]
39. Zhan, E. Research on Vision Based Algorithm for Detecting Vacant Parking Spaces. Master's Thesis, Dalian University of Technology, Dalian, China, 2021. [CrossRef]
40. Zhao, K.; Han, Q.; Zhang, C.B.; Xu, J.; Cheng, M.M. Deep hough transform for semantic line detection. *IEEE Trans. Pattern Anal. Mach. Intell.* **2021**, *44*, 4793–4806. [CrossRef]
41. Zhu, W.; Huang, H.; Ma, J. Research on parking space detection algorithm based on image recognition. *Automot. Eng.* **2019**, *41*, 744–749+756. [CrossRef]
42. Rahman, S.; Ramli, M.; Arnia, F.; Muharar, R.; Luthfi, M.; Sundari, S. Analysis and comparison of hough transform algorithms and feature detection to find available parking spaces. *J. Phys. Conf. Ser.* **2020**, *1566*, 012092. [CrossRef]
43. Lu, Y.; Shen, Y.; Wang, Y.; Guo, L. Research on parking space detection based on machine vision. *Pract. Automot. Technol.* **2023**, *48*, 56–60. [CrossRef]
44. Yan, Y. Research on Detection Method of Empty Parking Space and Relative Parking Position of Vehicle Body Based on Machine Vision. Master's Thesis, Jiangsu University, Zhenjiang, China, 2022. [CrossRef]
45. Li, C.; Fu, H.; Chen, G. Improved template matching fast recognition algorithm. *J. Henan Univ. Technol. (Nat. Sci. Ed.)* **2020**, *39*, 106–111. [CrossRef]

46. Ying, J.; Chen, W.; Yang, H.; He, S.; Liu, J. Research on parking space recognition and counting algorithm based on affine transformation and template matching. *Comput. Appl. Res.* **2022**, *39*, 919–924. [CrossRef]
47. Beijing University of Aeronautics and Astronautics. Recognition Method of Vacant Parking Space in Outdoor Parking Lot Based on Aerial Images. CN Patent CN201510562952.7, 24 July 2018.
48. Ma, S.; Fang, W.; Jiang, H.; Han, M.; Li, C. Parking space recognition method based on parking space feature construction in the scene of autonomous valet parking. *Appl. Sci.* **2021**, *11*, 2759. [CrossRef]
49. Xu, Y.; Zhou, J.; Zhou, Y. Stereo vision matching technology. *Comput. Eng. Appl.* **2003**, 1–5+62. [CrossRef]
50. Xiao, H.; Hu, Z.; Zhou, Z.; Wu, J. Rapid detection method of parking space occupation based on binocular back projection. *Automot. Eng.* **2021**, *43*, 650–656. [CrossRef]
51. Wu, Z.; Sun, W.; Wang, M.; Wang, X.; Ding, L.; Wang, F. Psdet: Efficient and universal parking slot detection. In Proceedings of the 2020 IEEE Intelligent Vehicles Symposium (IV), Las Vegas, NV, USA, 19 October–13 November 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 290–297.
52. Li, Z.; Lin, C.; Zhu, H.; Liao, S.; Bai, Y. Small target detection based on structure awareness and global context information. *Comput. Eng. Appl.* **2024**, *60*, 292–298.
53. Encío, L.; Díaz, C.; Del-Blanco, C.R.; Jaureguizar, F.; García, N. Visual Parking Occupancy Detection Using Extended Contextual Image Information via a Multi-Branch Output ConvNeXt Network. *Sensors* **2023**, *23*, 3329. [CrossRef] [PubMed]
54. Do, H.; Choi, J.Y. Context-based parking slot detection with a realistic dataset. *IEEE Access* **2020**, *8*, 171551–171559. [CrossRef]
55. Choi, R.Y.; Coyner, A.S.; Kalpathy-Cramer, J.; Chiang, M.F.; Campbell, J.P. Introduction to machine learning, neural networks, and deep learning. *Transl. Vis. Sci. Technol.* **2020**, *9*, 14. [PubMed]
56. Gutmann, S.; Maget, C.; Spangler, M.; Bogenberger, K. Truck parking occupancy prediction: Xgboost-LSTM model fusion. *Front. Future Transp.* **2021**, *2*, 693708. [CrossRef]
57. Gomari, S.; Domakuntla, R.; Knoth, C.; Antoniou, C. Development of a data-driven on-street parking information system using enhanced parking features. *IEEE Open J. Intell. Transp. Syst.* **2023**, *4*, 30–47. [CrossRef]
58. Ionita, A.; Pomp, A.; Cochez, M.; Meisen, T.; Decker, S. Where to park? Predicting free parking spots in unmonitored city areas. In Proceedings of the 8th International Conference on Web Intelligence, Mining and Semantics, Novi Sad, Serbia, 25–27 June 2018; pp. 1–12.
59. Wang, L. Research on Indoor Parking Space Detection Algorithm Based on Visual Analysis. Master's Thesis, Guizhou University for Nationalities, Guiyang, China, 2019.
60. Chen, T.; Li, Q. Parking space detection and tracking algorithm based on look around system. *Automot. Technol.* **2021**, 1–6. [CrossRef]
61. Li, K.; Wang, X.; Lin, H.; Li, L.; Yang, Y.; Meng, C.; Gao, J. Overview of single-stage small target detection methods in deep learning. *Comput. Sci. Explor.* **2022**, *16*, 41–58.
62. Xu, Y.; Li, J.; Dong, Y.; Zhang, X. Overview of Yolo series target detection algorithms. *Comput. Sci. Explor.* **2024**, *18*, 2221–2238.
63. Li, W.; Cao, L.; Yan, L.; Li, C.; Feng, X.; Zhao, P. Vacant parking slot detection in the around view image based on deep learning. *Sensors* **2020**, *20*, 2138. [CrossRef] [PubMed]
64. Xiao, A.; Doshi, D.; Wang, L.; Gorantla, H.; Heitzmann, T.; Groth, P. Parking spot classification based on surround view camera system. In Proceedings of the Applications of Machine Learning 2023, San Diego, CA, USA, 23–24 August 2023; SPIE: Bellingham, WA, USA, 2023; Volume 12675, pp. 129–136.
65. Zhang, B.; Wang, Y.; Pan, Z.; Wang, Y.; Yang, C.; Wang, X. Research on single-stage multi task Yolo parking algorithm based on Obstacle and parking space detection. *J. Hefei Univ. Technol. (Nat. Sci. Ed.)* **2024**, *47*, 1–6+61.
66. Chen, D.; Sun, S.; Wang, Y.; Lei, Z. Research on small target detection based on improved SSD algorithm. *Sens. Microsyst.* **2023**, *42*, 65–68+72. [CrossRef]
67. Zhang, B.; Wang, Z.; Li, S. Research on parking angle detection algorithm based on improved convolutional neural network. *Sens. Microsyst.* **2023**, *42*, 62–65+69. [CrossRef]
68. Li, H.; Zhang, H.; Wang, M.; Shen, B. Intelligent parking space detection based on deep learning. *J. China Acad. Electron. Sci.* **2021**, *16*, 1264–1269.
69. Min, C.; Xu, J.; Xiao, L.; Zhao, D.; Nie, Y.; Dai, B. Attentional graph neural network for parking-slot detection. *IEEE Robot. Autom. Lett.* **2021**, *6*, 3445–3450. [CrossRef]
70. Suhr, J.K.; Jung, H.G. End-to-end trainable one-stage parking slot detection integrating global and local information. *IEEE Trans. Intell. Transp. Syst.* **2021**, *23*, 4570–4582. [CrossRef]
71. Guo, S. Deep Learning Parking Space Recognition Based on Look Around Image. Master's Thesis, Xihua University, Chengdu, China, 2023.
72. Shen, X.; Liu, T.; Wang, Y.; Liu, J. Parking space state detection algorithm based on convolutional network structure re parameterization. *J. Jilin Univ. (Eng. Ed.)* **2022**, *52*, 2898–2905. [CrossRef]

73. Zhao, Z.; Gong, J. Multi stage target detection model based on candidate box multi-step iterative optimization. *China New Commun.* **2021**, *23*, 50–55.
74. Wang, X.; Zhu, X.; Yao, M. Target detection method based on improved faster RCNN. *High Tech Commun.* **2021**, *31*, 489–499.
75. Bui, Q.H.; Suhr, J.K. CNN-based two-stage parking slot detection using region-specific multi-scale feature extraction. *IEEE Access* **2023**, *11*, 58491–58505. [CrossRef]
76. Karanam, M.; Reddy, A.K.; Mathyam, P.; Mohammed, A.; Naik, S. Automatic Vacant Parking Places Management System Using Multicamera Vehicle Detection. *E3s Web Conf.* **2023**, *391*, 01045. [CrossRef]
77. Huang, C.; Yang, S.; Luo, Y.; Wang, Y.; Liu, Z. Visual detection and image processing of parking space based on deep learning. *Sensors* **2022**, *22*, 6672. [CrossRef] [PubMed]
78. Marek, M. Image-based parking space occupancy classification: Dataset and baseline. *arXiv* **2021**, arXiv:2107.12207.
79. Gao, C.; Xu, Z.; Wu, D.; Yu, C.; Sang, N. Overview of deep learning real-time semantic segmentation. *Chin. J. Image Graph.* **2024**, *29*, 1119–1145.
80. Jian, D.H.; Lin, C.H. Vision-based parking slot detection based on end-to-end semantic segmentation training. In Proceedings of the 2020 IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 4–6 January 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 1–4.
81. Zambanini, S.; Loghin, A.M.; Pfeifer, N.; Soley, E.M.; Sablatnig, R. Detection of parking cars in stereo satellite images. *Remote Sens.* **2020**, *12*, 2170. [CrossRef]
82. Lee, S.; Lim, W.; Sunwoo, M. Robust parking path planning with error-adaptive sampling under perception uncertainty. *Sensors* **2020**, *20*, 3560. [CrossRef] [PubMed]
83. Lai, C.; Yang, Q.; Guo, Y.; Bai, F.; Sun, H. Semantic segmentation of panoramic images for real-time parking slot detection. *Remote Sens.* **2022**, *14*, 3874. [CrossRef]
84. Li, B. Research on Parking Space Detection Based on Deep Learning. Master's Thesis, Dalian University of Technology, Dalian, China, 2023.
85. Huang, T.; Li, H.; Zhou, G.; Li, S.; Wang, Y. Overview of case segmentation methods. *Comput. Sci. Explor.* **2023**, *17*, 810–825.
86. Zhou, X.; Huang, D. Overview of small sample instance segmentation. *J. East China Norm. Univ. (Nat. Sci. Ed.)* **2022**, 136–146. [CrossRef]
87. Cheng, Y.; Yu, J. Intelligent recognition algorithm of parking space state based on mask r-cnn. *Comput. Simul.* **2021**, *38*, 236–239+471.
88. Naufal, A.; Fatichah, C.; Suciati, N. Preprocessed mask RCNN for parking space detection in smart parking systems. *Int. J. Intell. Eng. Syst.* **2020**, *13*, 255–265. [CrossRef]
89. Jiang, S.; Jiang, H.; Ma, S.; Jiang, Z. Detection of parking slots based on mask R-CNN. *Appl. Sci.* **2020**, *10*, 4295. [CrossRef]
90. Li, X. Research on 3D Target Detection Technology Based on Key Points. Master's Thesis, Southwest Jiaotong University, Chengdu, China, 2022.
91. Wang, Z.; Sang, H. End to end parking space detection network integrating parking line direction. *J. Electron. Meas. Instrum.* **2024**, *38*, 86–93. [CrossRef]
92. Qin, X.; Yin, Z.; Zhang, S.; Huang, S.; Zhang, R.; Lu, T.; Liu, S.; Hu, M. End to end parking space detection method based on panoramic view. *J. Hunan Univ. (Nat. Sci. Ed.)* **2024**, *51*, 177–186. [CrossRef]
93. Shan, K.; Sang, H. Research on parking space detection technology based on marker point feature under panoramic video. *J. Electron. Meas. Instrum.* **2022**, *36*, 203–210. [CrossRef]
94. Shan, K. Research on Parking Space Detection Technology Based on Key Point Features in Panoramic Video. Master's Thesis, Shenyang University of Technology, Shenyang, China, 2022.
95. Yue, P. Research on Parking Space Detection Method in Panoramic Image Based on Key Point Detection. Master's Thesis, Hunan University, Changsha, China, 2023.
96. InBev Supercomputing (Nanjing) Technology Co., Ltd. A Parking Space Detection Method Based on Parking sPace Key Point Instance Detection. CN Patent CN202210130169.3, 10 June 2022.
97. Sun, R. Optimization for deep learning: Theory and algorithms. *arXiv* **2019**, arXiv:1912.08957.
98. Li, W.; Zhong, J.; Sun, H.; Li, B.; Tang, Q.; Zheng, Z. An empty parking space detection method based on look around image. *J. Dalian Univ. Technol.* **2022**, *62*, 535–542.
99. Hayawi, D.; Muhammad, S.S. Detection parking Spaces by using the ResNet50 Algorithm. *J. Al-Qadisiyah Comput. Sci. Math* **2022**, *14*, 11–21. [CrossRef]
100. Nurullayev, S.; Lee, S.W. Generalized parking occupancy analysis based on dilated convolutional neural network. *Sensors* **2019**, *19*, 277. [CrossRef] [PubMed]
101. Nyambal, J.; Klein, R. Automated parking space detection using convolutional neural networks. In Proceedings of the 2017 Pattern Recognition Association of South Africa and Robotics and Mechatronics (PRASA-RobMech), Bloemfontein, South Africa, 29 November–1 December 2017; IEEE: Piscataway, NJ, USA, 2017; pp. 1–6.

102. Martynova, A.; Kuznetsov, M.; Porvatov, V.; Tishin, V.; Kuznetsov, A.; Semenova, N.; Kuznetsova, K. Revising deep learning methods in parking lot occupancy detection. *arXiv* **2023**, arXiv:2306.04288.
103. Wang, J.; Zhang, A. Parking space status detection of open parking lot based on overhead video image. *Mech. Des. Manuf.* **2023**, 67–70. [CrossRef]
104. Siddiqui, S.Y.; Khan, M.A.; Abbas, S.; Khan, F. Smart occupancy detection for road traffic parking using deep extreme learning machine. *J. King Saud-Univ.-Comput. Inf. Sci.* **2022**, *34*, 727–733.
105. Yuldashev, Y.; Mukhiddinov, M.; Abdusalomov, A.B.; Nasimov, R.; Cho, J. Parking lot occupancy detection with improved MobileNetV3. *Sensors* **2023**, *23*, 7642. [CrossRef] [PubMed]
106. Manh, H.N. Rethinking Feature Generalization in Vacant Space Detection. *Sensors* **2023**, *23*, 4776. [CrossRef] [PubMed]
107. Amato, G.; Carrara, F.; Falchi, F.; Gennaro, C.; Meghini, C.; Vairo, C. Deep learning for decentralized parking lot occupancy detection. *Expert Syst. Appl.* **2017**, *72*, 327–334. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.